by oxen on carts or transported in pieces strapped to the backs of camels. The light artillery was pulled by horses. The very largest cannon, required for the siege of stubborn strongholds, would be manufactured on the spot. Although siege warfare was the primary use of artillery, its effectiveness was limited. The Mughals continued to resort to traditional tactics such as digging mines under fortress walls and simply starving the defenders into submission. The main impact of cannon seems to have been psychological, adding to the mounting pressure on besieged forces to agree surrender terms.

ON THE BATTLEFIELD

Mughal forces typically took form for battle with massed light cavalry on the flanks and heavy cavalry in the centre. The artillery and the musket-armed infantry were deployed as a defensive block, with a line of war-elephants behind them. The presence of gunpowder weapons rendered elephants useless as an offensive force, since the noise and smoke made them panic uncontrollably. They functioned instead as command and observation posts, and as a last line of defence. The psychological impact of their towering presence was also significant.

The Mughals would usually start a battle by sending their mounted archers forward on the flanks to deluge the enemy with arrows. After this highly effective softening-up phase, the heavy cavalry would charge and engage the opposition in a general mêlée. An alternative battle plan, executed successfully against numerically superior forces at Panipat and in several other conflicts, was for the Mughal cavalry to feign a retreat, drawing the enemy into the fire of massed cannon and infantry firearms. The cavalry would then counterattack to devastating effect.

MUGHAL WEAKNESSES

Although the Mughal Empire reached its zenith, territorially, under Aurengzeb (ruled 1658–1707), it was by then in military decline. Mughal armies suffered defeats at the hands of the Marathas in southern India from the late 17th century and had to make peace with them. They were also defeated by Persian and Afghan armies in the 1730s – led by Nadir Shah, ruler of Persia – well before the British takeover of India got under way. Shah invaded Delhi, carrying away many treasures, including the Peacock Throne in 1739.

The Mughal forces were in many ways highly sophisticated. Their engineers, for example, were skilled road builders who opened up routes for the army to use through otherwise impassable terrain. But they were limited in their adoption and use of new military technology. Their cavalry did not carry pistols and their infantry had neither pikes nor bayonets. They failed to move on from the matchlock to the superior flintlock musket, still employing the old-fashioned firearm into the 18th century. Despising infantry, they did not subject them to drill, so that foot soldiers were slow to load their muskets and could not fire in volleys.

The fragility of Mughal power was, however, largely a consequence of the army's fundamental structure, depending on warlords bringing their retainers to serve the emperor. This prevented the creation of any permanent hierarchy of command, since each warlord owed allegiance directly to the ruler and accepted orders only from him. Amirs were inevitably tempted to assert their independence in their provincial power base, rather than joining the imperial army. Professional soldiers would sell their services to another employer once imperial wealth and power waned. The Mughal Empire theoretically continued to exist until 1857, but by then it was a mere ghost of its former glory.

However, the empire's legacy was long-lasting; many monuments of the Mughal period (most famously the Taj Mahal) are still in use today.

Mughals in India
Babur's defeat of Sultan Lodi of Delhi at Panipat in 1526 allowed him to establish Mughal rule in northern India. From this base, the Mughals expanded and consolidated territory to the south for almost a further 200 years, under successive rulers including Akbar and Aurengzeb.





Animal army
Heavily armoured elephants in battle were a fearsome sight for the Mughals' enemies. Furthermore, mounted warriors could exploit the animals' height to direct the battle.



Indian kard (dagger) and scabbard
Brought to India through Mughal expansion, this straight-bladed, single-edged *kard* was in use across much of the Islamic world by the 18th century.